Benjamin Diedrichsen

Polyglot Software Architect

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The design and construction of software systems - be it small libraries, frameworks or midsize software landscapes - is my job and one of my greatest passions. This includes many aspects of the software lifecycle:
planning and design, data and domain modeling, prototyping, deployment, production and maintenance.

Polyglot

I write code in various languages using combinations of object oriented and functional paradigm. @home on the JVM [Java, Scala, Groovy], I recently became an enthusiastic user of the JavaScript ecosystem - node.js, coffeescript, livescript, react.js

Everything is a model

I enjoy the creative task of deconstructing reality into appropriate models of objects, functions, data structures and information flows. I take pride in writing beautiful and efficient code - git clone my open source projects and have a look.

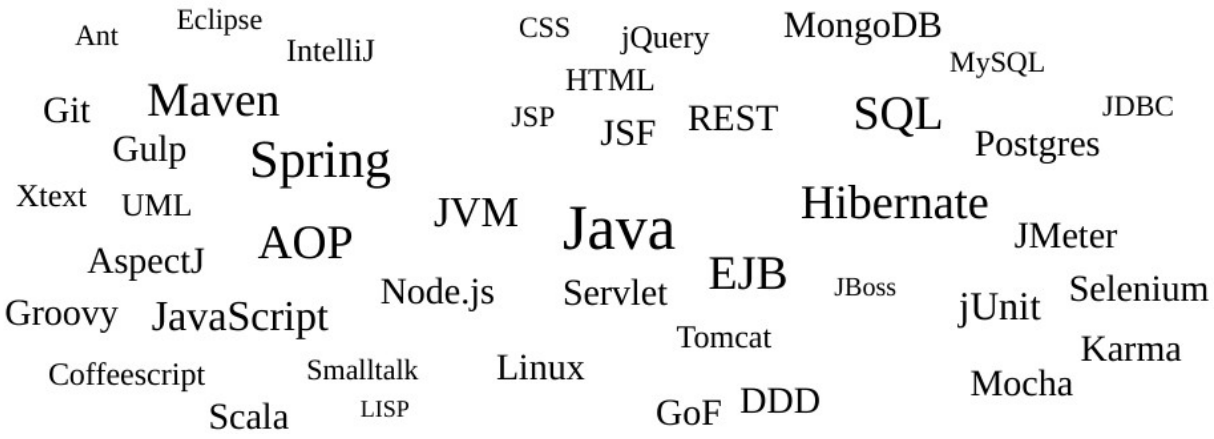
I also believe in the values of agile methodologies: open and honest communication, collective learning and team collaboration are as important as choosing the right technology stack.

You can hire me to

- design and prototype core modules or system architecture
- fix performance issues in application code or data models
- plan and implement major refactorings
- help with code reviews, technology decisions, development cycle and team process

I grant discounts if you

- have a an interesting technology stack
- solve social problems instead of consumer problems



Software projects (selection)

05/2016 – ongoing	**Rewrite of web/mobile app with Node.js backend**
www.mein-schulplan.de	Mobile and web app built on innovative stack using TFRP (reactjs+eventemitter) in frontend and Node.js, ES7 in backend.
Technologies	Node.js, ES7, coffeescript, reactjs, PouchDB, CouchDB, koa
Tasks	➢ Design and prototyping of entire application stack ➢ Providing infrastructure for unit, integration and e2e tests
06/2015 – 12/2015	**Implementation of REST-API Stack**
Okotta GmbH *www.okotta.com*	Mobile App with REST data backend. Scalable data repositories secured with JSON Web Tokens.
Technologies	Scala, Scalatra, MongoDB, reactivemongo, JWT, Swagger, WireMock, reactjs, ES6
Tasks	➢ Requirements engineering and domain modeling ➢ Design and implementaion of REST-API stack ➢ Evaluation of persistence technologies ➢ Implementation of full stack integration testing ➢ Internal Data Tool based on ES6 and react.js
06/2014-12/2014	**REST API built on Google App Engine and Memcache**
Small Improvements *small-improvements.com*	Web application running on GAE with BigTable and Memache. Angular frontend connects via Ajax to data backend.
Technologies	Java 6, Groovy, GAE, Memcache, Google Guava, Jackson, Jmeter, Spock, Gradle
Tasks	➢ Performance optimization of object serialization ➢ (Re-)Implementation of REST-API with composable serializers ➢ Integration of caching layer using Guava Cache ➢ Refactoring to patterns

03/2013 - 09/2013	**Automated provisioning of Virtual Machines in Data Centers**
ProfitBricks GmbH *www.profitbricks.com*	J2EE application handling customer requests for virtualized server infrastructure
Technologies	Java 7, EJB 3, JPA 2 (EclipseLink), Postgres, Glassfish, jUnit, cglib, OSGI, H2, Jenkings, Maven, git
Tasks	➢ Rewrite test infrastructure with internal DSL and H2 ➢ Lazy assertions with cglib and dynamic proxies ➢ Profiling and performance optimizations ➢ Code reviews and refactorings
02/2012 - 12/2012	**Web application with scalable relational social graph**
Alecto GmbH *www.friendsurance.de*	Online platform offering comparison services around insurances and a social graph for customers to form trust networks
Technologies	Java 7, EJB 3, JPA 2 (Hibernate), Spring 3, Postgres, Tomcat 7, Glassfish, Selenium, JSF, Maven, git
Tasks	➢ Full migration from EJB 3 (Glassfish) to Spring (Tomcat) ➢ Redesign of persistence abstraction layer ➢ Historized social graph based on immutable data (Postgres) ➢ Caching layer with synchronization across machines ➢ Selenium test infrastructure on grid server

Co-operative course of studies

15.10.2007 – 30.09.2008	**Softwaredevelopment ottogroup retail systems**
www.ottogroup.com	Support the migration of the ORM layer in a large J2EE landscape
Technologies	Java 6, JDO 2.0, JPA 1.0, jUnit, Spring, Xtext, Xtend,Xpand,
Tasks	➢ Extension of existing persistence infrastructure for operating JDO and JPA in parallel (soft migration strategy) ➢ Automatic transformation of all OR-mapping definitions as part of my diploma thesis

Degree program

01.10.2004 –30.09.2008	**Co-operative degree program 'Economics and computer science (Dipl. FH)' in co-operation with ottogroup**
Focue areas	Model driven software development ■ Modern software technologies ■ Datastream Managementsystems
Diploma thesis	Implementation of a generic transformation component for XML based models and its application to a model migration problem
Final grades	1.0 (thesis), 1.5 (diploma)

IT-Qualifications

- 8+ years software development experience in J2EE business environment
- Strong experience running J2EE technologies in production: EJB, Spring, JTA, JPA, Servlet, JSP, application servers, servlet containers
- Object relationial persistence: Data and object modeling, query optimization, Postgres, MySql
- Good understanding of architecture and design patterns: SOLID, Messaging, Pipes&Filters, REST, GoF Patterns
- Experience with model driven software development and modeling: Xtext, Xtend, Xpand, EMF, UML, AndroMDA
- Solid experience with XML technologies: XSD, XSLT, XPath
- Understanding of functional programming (Lisp/Scheme)
- Development Tools: Maven, Git, Ant, SVN,CVS,Jenkins
- Experienced Linux user

More

[github:mbassador] Java Event Bus with larger user base: HighPerformance implementation of publishsubscribe pattern. Low memory footprint, nonblocking iterators, high throughput

[github:mailody] Online trash mail service built with Node.js, MongoDB, restify, knockout.js, HTML 5, Java client for Selenium

[github:kermit] Kermit is an elegant, stream-based, extensible web scraper system. It is packed with pluggable features and offers a plugin mechanism for easy extension. Written in Node.js and coffeescript with NeDB as datastore.

Languages

*Foreign**

	oral	writing
English	C1	C2
Spanish	B2	C1
French	A1	A2

** Self evaluation according to refernce framework of European Language Portfolio (see appendix or www.coe.int/portfolio)*

Native Deutsch